

Metadata – Trunk Hive – Series « 2024_06_20_Csv »

Datasets: 2024_06_20_Environment.csv, 2024_06_20_LogHive.csv

Apiary location: Mas Lafont, 30170 CROS, France

Hive type: Empty wooden Log Hive.

Hive environment: open area

Description:

Log Hive, chestnut wood, covered by 2cm thick stone.



Figure 1: Trunk hive

List of sensors:

- 1 sensor installed on the overframe (central position) called "overframe"
- 1 sensor in upper hive body (under the overframe) called "top"
- 1 sensor in lower part of the hive body (at the brick layer surface) called "bot"
- 1 sensor below the hive (between the ground and the brick layer) called "below"
- 1 sensor in below the ground surface at 10cm called "ground10"
- 1 sensor in below the ground surface at 10cm called "ground20"



Environment:

- The luminosity (**6 sensors**) values in each direction (north, south, east, west, top, bottom) is measured on each face of a Dadant close (3m) to the trunk hive.
- Outside air sensor
- Wind speed

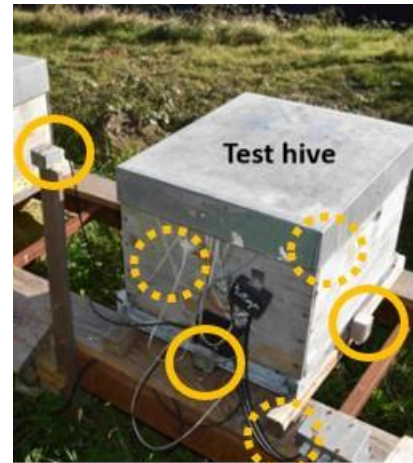


Figure 2: Luminosity sensors

Communications between the acquisition system and the sensors are done via the I2C protocol. The sensors provide an internal analog-to-digital conversion. The sensors were individually calibrated to cancel the bias error between them to allow more precise temperature difference measurements ($<0.5^{\circ}\text{C}$). Absolute temperature errors are probably around 1°C . The intercalibration were achieved by putting all sensors together in an insulated box.

The acquisition period is set to one minute except for the wind speed data which are acquired every 2 s and averaged over 1 min.

Data are stored using a csv format in 2 different files:

- 1) 2024_06_20_Environment.csv
- 2) 2024_06_20_LogHive.csv

Comment lines are prefixed with hash character '#'. Column titles are built using comma separated identifier:

identifier1[.identifier2[.identifier3...]].field

For example, the roof temperature of test hive is "trunk.roof.T".

Field name	Description	Unit	Sensor type	Sensor location	
Timestamp	Timestamp of measurement: Year/month/ Day/Hour/ min/sec	(local time)	PCF8523 (Real-time clock)+ DS3231 (Real-time clock module)	Acquisition system	
Epoch	Timestamp of measurement	Day since 1 January 0			
Wind.speed [m/s]	Wind speed	m/s	Sparkfun meteo system	Close to the apiary, 2m high	
Ground.T [deg]	Temperature at the ground surface ($\approx 1\text{cm}$)	$^{\circ}\text{C}$	NCT75	On the ground in a shaded area.	
Sky.T [deg]	Equilibrium temperature of a horizontal black steel plane exposed to sunlight and insulated at the rear face	$^{\circ}\text{C}$	NCT75	Fixed to the weather station	
Luminosity.south.als [lux]	Luminosity south	Lux	VEML7700 (als channel)	South side of the Dadant hive	
Luminosity.south.white [lux]	Luminosity south	Lux	VEML7700 (white channel)	South side of the Dadant hive	
Luminosity.top.als [lux]	Luminosity top	Lux	VEML7700 (als channel)	Top side of the Dadant hive	
Luminosity.top.white [lux]	Luminosity top	Lux	VEML7700 (white channel)	Top side of the Dadant hive	
Luminosity.north.als [lux]	Luminosity north	Lux	VEML7700 (als channel)	North side of the Dadant hive	
Test.luminosity.north.white [lux]	Luminosity north	Lux	VEML7700 (white channel)	North side of the Dadant hive	
Luminosity.west.als [lux]	Luminosity west	Lux	VEML7700 (als channel)	West side of the Dadant hive	

Luminosity.west.white [lux]	Luminosity west	Lux	VEML7700 (white channel)	West side of the Dadant hive	
Luminosity.below.als [lux]	Luminosity below	Lux	VEML7700 (als channel)	Downward facing below the Dadant hive	
Luminosity.below.white [lux]	Luminosity below	Lux	VEML7700 (white channel)	Downward facing below the Dadant hive	
Luminosity.east.als [lux]	Luminosity east	Lux	VEML7700 (als channel)	East side of the Dadant hive	
Luminosity.east.white [lux]	Luminosity east	Lux	VEML7700 (white channel)	East side of the Dadant hive	
Luminosity.horizontal.ALS [lux]	Luminosity horizontal plane	Lux	VEML7700 (als channel)	Upward facing fixed on the weather station	
Luminosity.horizontal.WHITE [lux]	Luminosity horizontal plane	Lux	VEML7700 (white channel)	Upward facing fixed on the weather station	
Luminosity.45.C H0 [lux]	Luminosity on a 45-tilted plane (azimuth=south)	Lux	TSL2561	Fixed on the weather station	
Luminosity.45.C H1 [lux]	Luminosity on a 45-tilted plane (azimuth=south)	Lux	TSL2561	Fixed on the weather station	
Luminosity.45.LUX [lux]	Luminosity on a 45-tilted plane (azimuth=south)	Lux	TSL2561	Fixed on the weather station	
Log.body.top.HR [%]	Humidity in the upper part of log hive	%	HIH8100	Inside hive body (top), in the centre	
Log.body.top.T [deg]	Temperature in the upper part of log hive	°C	HIH8100	Inside hive body (top), in the centre	
Log.body.bot.T [deg]	Temperature in the lower part of log hive	°C	HIH8100	Inside hive body (bottom), in the centre	

Log.body.bot.HR [%]	Humidity in the lower part of log hive	%	HIH8100	Inside hive body (bottom), in the centre	
Log.overframe.T [deg]	Temperature in the overframe part	°C	NCT75	Centre of the overframe	
Log.below.T [deg]	Temperature below the log hive	°C	NCT75	Between the ground and the brick layer	
Log.ground.10.T [deg]	Ground temperature below the log hive	%	HIH8100	10cm under the ground surface under the hive	
Log.ground.20.T [deg]	Ground temperature below the log hive	%	HIH8100	20cm under the ground surface under the hive	